

Phase 6 — Incident Response and ITSM

Harrington Capital plc | Infrastructure Engineer Simulation

Objective

Work a simulated P1 incident from detection to RCA. The incident: HC-APP01 disk at 97% — application returning 500 errors. File the overdue RCA in GLPI. Establish the incident management process.

By the end of this phase you will have: - A GLPI P1 incident ticket worked end-to-end - The incident resolved (disk freed) - RCA document filed and committed to `/incident-reports` - Change request raised for permanent fix

THE SCENARIO

It is Monday morning, Week 7.

You receive this message from James Okafor:

"We've got alerts firing on HC-APP01. The Flask app is returning 500s. Users in Finance can't access the ops portal. Can you look at this now? This is our third disk incident in two months — I need an RCA filed today and a change request through the CAB by Friday. Ticket number: INC-20260602-001."

BUILD

Step 1 — Open the GLPI incident ticket

Log into GLPI and create/update ticket INC-20260602-001:

- Title: `[P1] HC-APP01 disk full — application returning 500 errors`
- Priority: P1
- Category: Infrastructure / Storage

- Status: In Progress
- Assign to: yourself
- Description: paste the message from James

Screenshot the ticket at this stage.

Step 2 — Triage the incident

SSH into HC-APP01:

```
# Check disk usage
df -h

# Find what is consuming disk
du -sh /* 2>/dev/null | sort -rh | head -20
du -sh /var/log/* 2>/dev/null | sort -rh | head -10

# Check Docker
docker system df

# Check application logs
docker logs hc-app --tail=50
docker logs hc-nginx --tail=20
```

Document everything you find in the GLPI ticket.

Step 3 — Identify root cause

The disk is filled by one of these common causes:

Most likely: Docker logs or container volumes

```
# Check Docker overlay storage
du -sh /var/lib/docker/overlay2/* 2>/dev/null | sort -rh | head -10

# Check log sizes
ls -lh /var/lib/docker/containers/*/*.log 2>/dev/null
```

Or: Application logs not rotated

```
ls -lh /var/log/
find /var/log -name "*.log" -size +100M 2>/dev/null
```

Step 4 — Contain: free the disk

```
# Prune Docker (remove stopped containers, unused images, build cache)
docker system prune -f

# Rotate and compress large logs
sudo find /var/log -name "*.log" -size +50M -exec gzip {} \;

# Clear old journal entries
sudo journalctl --vacuum-size=500M

# Verify disk freed
df -h
```

Disk usage must drop below 75% before proceeding.

Step 5 — Verify application restored

```
curl -s http://localhost:8080/health
curl -s http://localhost:8080/db-check
```

Both must return status 200 / `"status": "healthy"`.

Step 6 — Update and close GLPI ticket

- Paste `df -h` output (before and after)
- Paste `curl /health` response
- Set status: **Solved**
- Set resolution: "Freed 8.2 GB by pruning Docker build cache and compressing rotated logs"
- Set resolution time

Screenshot the resolved ticket.

Step 7 — Write the RCA

Copy `/incident-reports/INC-TEMPLATE-RCA.md` and rename to: `incident-reports/INC-20260602-001-RCA.md`

Complete every section. Key fields:

Field	Value
Incident ID	INC-20260602-001
Priority	P1
Detection time	09:14 Monday (simulate)
Resolution time	10:47 Monday (simulate)
Duration	1 hour 33 minutes
Systems affected	HC-APP01
Users affected	Finance team (~12 users)

Root cause: Docker build cache and unrotated application logs consumed available disk space on HC-APP01 at a rate of approximately 2 GB per week. No disk usage alerting threshold was configured in Prometheus/Grafana.

Remediation: - Immediate: `docker system prune` and log compression - Short-term (7 days): Configure Grafana alert at 75% disk usage - Long-term (30 days): Implement Docker log rotation policy via `daemon.json`

Step 8 — Raise a change request

Create a second GLPI ticket: - Title: `CHG-20260602-001 – Implement disk usage alerting and Docker log rotation` - Type: Change Request - Priority: P2 - Reference RCA: INC-20260602-001

Step 9 — Commit

```
git add incident-reports/INC-20260602-001-RCA.md
git commit -m "docs(incidents): P1 RCA filed - INC-20260602-001 disk full HC-APP01 - Phase 6"
git push origin main
```

VERIFY

```
# Confirm disk healthy
df -h | grep -E "^/dev|Filesystem"

# Confirm app running
curl -s http://localhost:8080/health

# Confirm Docker cleaned up
docker system df
```

SUBMIT

#	Evidence	How to get it
1	GLPI ticket screenshot (In Progress)	Ticket showing P1 status and your triage notes
2	GLPI ticket screenshot (Solved)	Resolved ticket with resolution notes
3	<code>df -h</code> before screenshot	Shows >90% usage
4	<code>df -h</code> after screenshot	Shows <75% usage
5	<code>curl /health</code> output	JSON showing <code>"status": "healthy"</code>
6	RCA document GitHub URL	<code>incident-reports/INC-20260602-001-RCA.md</code>

Anti-fake check: GLPI ticket must show a real ticket number (system-assigned, e.g. Ticket #4). Screenshot must show the number.